

frequency range depending on their hardware capabilities.

The key deployments of SDR include signal intelligence, spectrum monitoring, satellite signal reception, IoT protocol analysis, amateur radio experimentation, GSM and LTE research, 5G, 6G, weather satellite decoding, aircraft tracking (ADS-B) and academic teaching in digital signal processing. Government agencies and defence units can use SDR for surveillance, spectrum management and security applications. The telecom and corporate industries can use SDR for network testing, interference analysis and prototyping of wireless communication systems. In the education sector and academics, SDR platforms are widely used to demonstrate modulation techniques, filtering, Fourier transforms, and real-time signal visualisation.

Modules and programming platforms for SDR

A variety and number of SDR hardware modules are available including USB-based receivers such as RTL-SDR Blog V4. These are commonly used for AM/FM/shortwave reception mapping, ADS-B decoding, analytics and satellite monitoring. Advanced platforms for transceivers include USRP and HackRF, which are used for research and intelligence-based experiments including wireless security testing and protocol development.

Popular software platforms and tools like GNU Radio provide graphical user interface (GUI) and Python-based programming environments for building SDR applications. Tools like SDR#, SDR++ and GQRX enable real-time spectrum visualisation with signal decoding.

Programming platforms and languages such as C++, Python, Octave and MATLAB are widely used for implementing modulation, demodulation, signal intelligence schemes, FFT analysis, filtering



Figure 1: Key use cases of SDR

Table 1: Modules, devices, kits and programming platforms for SDR

Module/Platform	Primary use	Type
KiwiSDR	Remote HF monitoring	Cloud-based SDR
RTL-SDR Blog V4	Low-cost RF reception	Hardware (Receiver)
HackRF One	RF experimentation	Hardware (Transceiver)
USRP	Advanced wireless research	Hardware (Research SDR)
LimeSDR Mini	LTE/5G prototyping	Hardware (Transceiver)
BladeRF	FPGA-based development	Hardware (Transceiver)
GNU Radio	DSP flowgraph development	Software framework
SDR#	Real-time spectrum analysis	Software application
MATLAB	Signal processing research	Programming platform
SDR++	Multi-platform SDR control	Software application
Airspy Mini	Wideband signal monitoring	Hardware (Receiver)
GQRX	Spectrum visualisation	Software application
CubicSDR	Cross-platform SDR interface	Software application
HDSDR	Advanced RF analysis	Software application
SoapySDR	Multi-device integration	SDR middleware
Python	SDR automation and AI	Programming language
LabVIEW	Instrument control	Programming platform
OpenWebRX	Browser-based reception	Web SDR platform
Pothos SDR	Modular DSP development	Software framework
SDR Console	Wideband spectrum monitoring	Software application